

Сетевые технологии

Лабораторная работа №6

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Цель данной работы — изучение принципов распределения и настройки адресного пространства на устройствах сети.

Исходная сеть 172.16.20.0/24	
Характеристика	Значение
Адрес сети	172.16.20.0/24
Длина префикса	24 бита
Маска	255.255.255.0
Broadcast-адрес	172.16.20.255/24
Адрес сети в двоичной форме	10101100 00010000 00010100 00000000
Маска в двоичной форме	11111111 11111111 11111111 00000000
Broadcast-адрес в двоичной форме	10101100 00010000 00010100 11111111
Число возможных подсетей (при /30)	$2^6 = 64$

Рис. 1: Задание 1.1

Исходная сеть 10.10.1.64/26	
Характеристика	Значение
Адрес сети	10.10.1.64/26
Длина префикса	26 бит
Маска	255.255.255.192
Broadcast-адрес	10.10.1.127/26
Адрес сети в двоичной форме	00001010 00001010 00000001 01000000
Маска в двоичной форме	11111111 11111111 11111111 11000000
Broadcast-адрес в двоичной форме	00001010 00001010 00000001 01111111
Число возможных подсетей (при /30)	$2^4 = 16$

Рис. 2: Задание 1.2

Исходная сеть 10.10.1.0/26	
Характеристика	Значение
Адрес сети	10.10.1.0/26
Длина префикса	26 бит
Маска	255.255.255.192
Broadcast-адрес	10.10.1.63/26
Адрес сети в двоичной форме	00001010 00001010 00000001 00000000
Маска в двоичной форме	11111111 11111111 11111111 11000000
Broadcast-адрес в двоичной форме	00001010 00001010 00000001 00111111
Число возможных подсетей (при /30)	$2^4 = 16$

Рис. 3: Задание 1.3

Исходная сеть	
Характеристика	Значение
Адрес сети	2001:db8:c0de::/48
Длина префикса	48 бит
Маска (префикс)	/48
Broadcast-адрес	не применимо (IPv6)
Адрес сети в двоичной форме	00100000 00000001 : 00001101 10111000 : 11000000 11011110 : 00000000 00000000 : 00000000 00000000 : 00000000 00000000 : 00000000 00000000 : 00000000 00000000
Маска в двоичной форме	11111111 11111111 : 11111111 11111111 : 11111111 11111111 : 00000000 00000000 : 00000000 00000000 : 00000000 00000000 : 00000000 00000000 : 00000000 00000000
Broadcast-адрес в двоичной форме	не применимо
Число возможных подсетей (теоретическое)	2^{80} (при максимальном дроблении до /128)

Рис. 4: Задание 2.1

Исходная сеть	
Характеристика	Значение
Адрес сети	2a02:6b8::/64
Длина префикса	64 бита
Маска (префикс)	/64
Broadcast-адрес	не применимо
Адрес сети в двоичной форме	00101010 00000010 : 00000110 10110000 : 00000000 00000000 : 00000000 00000000 : 00000000 00000000 : 00000000 00000000 : 00000000 00000000 : 00000000 00000000
Маска в двоичной форме	11111111 11111111 : 11111111 11111111 : 11111111 11111111 : 11111111 11111111 : 00000000 00000000 : 00000000 00000000 : 00000000 00000000 : 00000000 00000000
Broadcast-адрес в двоичной форме	не применимо
Число возможных подсетей (теоретическое)	2^{64}

Рис. 5: Задание 2.2

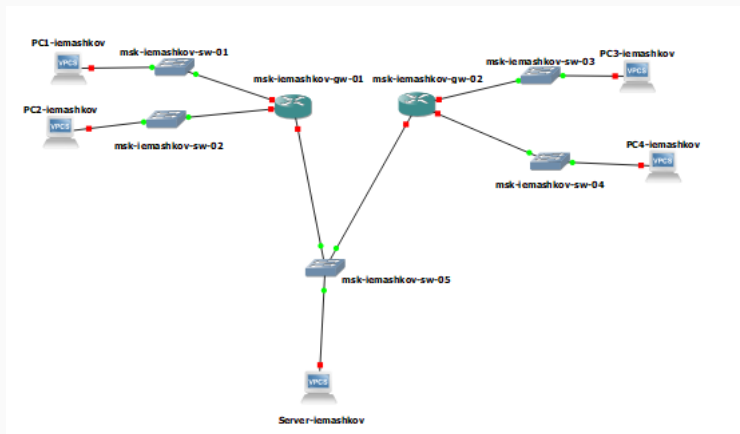
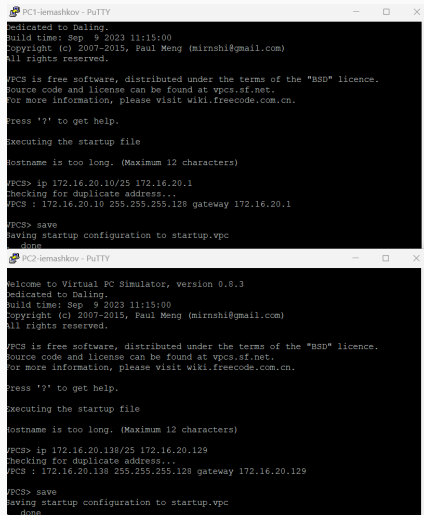


Рис. 6: Топология

Выполнение лабораторной работы



The image displays two screenshots of a PuTTY terminal window titled "PC1-iemashkov - PuTTY". The terminal shows the startup sequence of a Virtual PC Simulator, version 0.8.3, which is dedicated to Daling. The build time is Sep 9 2023 11:15:00, and the copyright is (c) 2007-2015, Paul Meng (mirnshi@gmail.com). The license is BSD, and the source code and license can be found at vpcs.sf.net. The terminal shows the execution of the startup file, which sets the hostname to "PC1" and the IP address to "172.16.20.10/25". The gateway is set to "172.16.20.1". The terminal also shows the execution of the "save" command, which saves the startup configuration to "startup.vpc".

```
PC1-iemashkov - PuTTY
dedicated to Daling.
Build time: Sep 9 2023 11:15:00
Copyright (c) 2007-2015, Paul Meng (mirnshi@gmail.com)
All rights reserved.

VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

Hostname is too long. (Maximum 12 characters)

VPCS> ip 172.16.20.10/25 172.16.20.1
Checking for duplicate address...
VPCS : 172.16.20.10 255.255.255.128 gateway 172.16.20.1

VPCS> save
Saving startup configuration to startup.vpc
done
```

The second screenshot shows the same terminal window, but with the IP address changed to "172.16.20.138/25". The gateway remains "172.16.20.129".

```
PC2-iemashkov - PuTTY
Welcome to Virtual PC Simulator, version 0.8.3
dedicated to Daling.
Build time: Sep 9 2023 11:15:00
Copyright (c) 2007-2015, Paul Meng (mirnshi@gmail.com)
All rights reserved.

VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

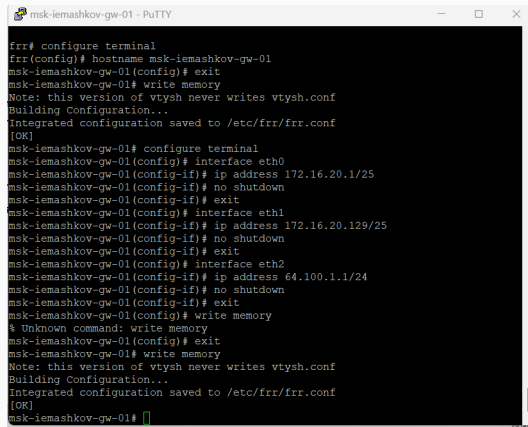
Hostname is too long. (Maximum 12 characters)

VPCS> ip 172.16.20.138/25 172.16.20.129
Checking for duplicate address...
VPCS : 172.16.20.138 255.255.255.128 gateway 172.16.20.129

VPCS> save
Saving startup configuration to startup.vpc
done
```

Рис. 7: Настройка ipv4 адресов на pc1 и pc2

Выполнение лабораторной работы



```
msk-iemashkov-gw-01 - PuTTY
frr# configure terminal
frr(config)# hostname msk-iemashkov-gw-01
msk-iemashkov-gw-01(config)# exit
msk-iemashkov-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-iemashkov-gw-01# configure terminal
msk-iemashkov-gw-01(config)# interface eth0
msk-iemashkov-gw-01(config-if)# ip address 172.16.20.1/25
msk-iemashkov-gw-01(config-if)# no shutdown
msk-iemashkov-gw-01(config-if)# exit
msk-iemashkov-gw-01(config)# interface eth1
msk-iemashkov-gw-01(config-if)# ip address 172.16.20.129/25
msk-iemashkov-gw-01(config-if)# no shutdown
msk-iemashkov-gw-01(config-if)# exit
msk-iemashkov-gw-01(config)# interface eth2
msk-iemashkov-gw-01(config-if)# ip address 64.100.1.1/24
msk-iemashkov-gw-01(config-if)# no shutdown
msk-iemashkov-gw-01(config-if)# exit
msk-iemashkov-gw-01(config)# write memory
% Unknown command: write memory
msk-iemashkov-gw-01(config)# exit
msk-iemashkov-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-iemashkov-gw-01#
```

Рис. 8: Настройка первого маршрутизатора на ipv4 адреса

Выполнение лабораторной работы

```
PC1-iemashkov - PuTTY
Copyright (c) 2007-2015, Paul Meng (mirnshi@gmail.com)
All rights reserved.

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Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

Checking for duplicate address...
VPCS : 172.16.20.10 255.255.255.128 gateway 172.16.20.1

VPCS> ping 172.16.20.138

84 bytes from 172.16.20.138 icmp_seq=1 ttl=63 time=8.772 ms
84 bytes from 172.16.20.138 icmp_seq=2 ttl=63 time=3.716 ms
84 bytes from 172.16.20.138 icmp_seq=3 ttl=63 time=2.499 ms
84 bytes from 172.16.20.138 icmp_seq=4 ttl=63 time=2.266 ms
84 bytes from 172.16.20.138 icmp_seq=5 ttl=63 time=1.894 ms

PC2-iemashkov - PuTTY
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All rights reserved.

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Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

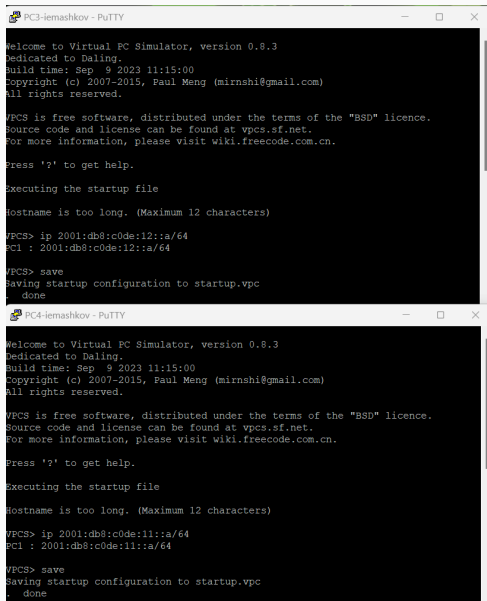
Executing the startup file

Checking for duplicate address...
VPCS : 172.16.20.138 255.255.255.128 gateway 172.16.20.129

VPCS> ping 172.16.20.10

84 bytes from 172.16.20.10 icmp_seq=1 ttl=63 time=3.026 ms
84 bytes from 172.16.20.10 icmp_seq=2 ttl=63 time=1.975 ms
84 bytes from 172.16.20.10 icmp_seq=3 ttl=63 time=2.099 ms
84 bytes from 172.16.20.10 icmp_seq=4 ttl=63 time=1.889 ms
84 bytes from 172.16.20.10 icmp_seq=5 ttl=63 time=1.594 ms
```

Выполнение лабораторной работы



The image displays two screenshots of a terminal window titled "PC3-lemashkov - PuTTY" and "PC4-lemashkov - PuTTY". Both windows show the same text, which is the output of the Virtual PC Simulator version 0.8.3. The text includes a welcome message, build time, copyright information, and a BSD license notice. It then shows the execution of a startup file, which results in an error: "Hostname is too long. (Maximum 12 characters)". Following this, the user enters the command "VPCS> ip 2001:db8:c0de:12::a/64" for PC1 and "VPCS> save", which saves the startup configuration to "startup.vpc".

```
PC3-lemashkov - PuTTY
Welcome to Virtual PC Simulator, version 0.8.3
Dedicated to Daling.
Build time: Sep  9 2023 11:15:00
Copyright (c) 2007-2015, Paul Meng (mirnshi@gmail.com)
All rights reserved.

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Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

Hostname is too long. (Maximum 12 characters)

VPCS> ip 2001:db8:c0de:12::a/64
PC1 : 2001:db8:c0de:12::a/64

VPCS> save
Saving startup configuration to startup.vpc
. done

PC4-lemashkov - PuTTY
Welcome to Virtual PC Simulator, version 0.8.3
Dedicated to Daling.
Build time: Sep  9 2023 11:15:00
Copyright (c) 2007-2015, Paul Meng (mirnshi@gmail.com)
All rights reserved.

VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

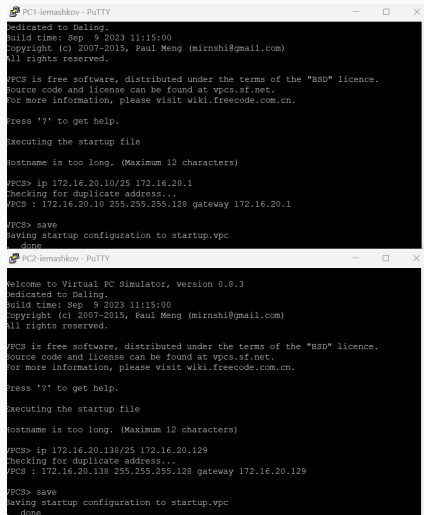
Executing the startup file

Hostname is too long. (Maximum 12 characters)

VPCS> ip 2001:db8:c0de:11::a/64
PC1 : 2001:db8:c0de:11::a/64

VPCS> save
Saving startup configuration to startup.vpc
. done
```

Выполнение лабораторной работы



The image displays two screenshots of PuTTY terminal windows, each showing the VPCS (Virtual PC Simulator) interface. The top window, titled 'PC1-iemashkov - PuTTY', shows the initial boot sequence and configuration for PC1. The bottom window, titled 'PC2-iemashkov - PuTTY', shows the same sequence for PC2. Both windows display the VPCS version 0.8.3, build time (Sep 9 2023 11:15:00), and copyright information (© 2007-2015, Paul Meng (mirnshi@gmail.com)). The configuration for PC1 includes setting the IP address to 172.16.20.10/25, the gateway to 172.16.20.1, and saving the configuration. The configuration for PC2 includes setting the IP address to 172.16.20.129/25, the gateway to 172.16.20.129, and saving the configuration. Both windows also show a warning about the hostname being too long (Maximum 12 characters).

```
PC1-iemashkov - PuTTY
dedicated to Daling.
Build time: Sep 9 2023 11:15:00
Copyright (c) 2007-2015, Paul Meng (mirnshi@gmail.com)
All rights reserved.

VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

Hostname is too long. (Maximum 12 characters)

VPCS> ip 172.16.20.10/25 172.16.20.1
Checking for duplicate address...
VPCS : 172.16.20.10 255.255.255.128 gateway 172.16.20.1

VPCS> save
Saving startup configuration to startup.vpc
done

PC2-iemashkov - PuTTY
Welcome to Virtual PC Simulator, version 0.8.3
dedicated to Daling.
Build time: Sep 9 2023 11:15:00
Copyright (c) 2007-2015, Paul Meng (mirnshi@gmail.com)
All rights reserved.

VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

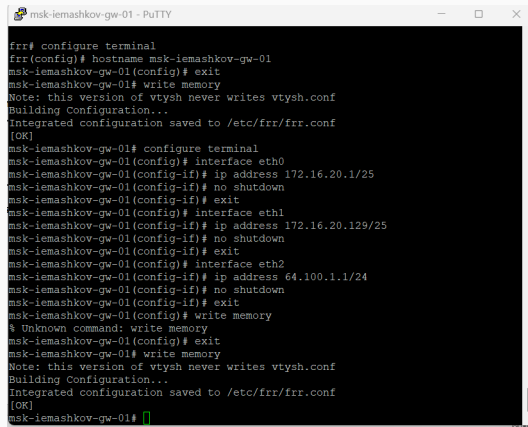
Hostname is too long. (Maximum 12 characters)

VPCS> ip 172.16.20.129/25 172.16.20.129
Checking for duplicate address...
VPCS : 172.16.20.129 255.255.255.128 gateway 172.16.20.129

VPCS> save
Saving startup configuration to startup.vpc
done
```

Рис. 11: Настройка ipv6 адресов на pc3 и pc4

Выполнение лабораторной работы



```
msk-iemashkov-gw-01 - PuTTY
frr# configure terminal
frr(config)# hostname msk-iemashkov-gw-01
msk-iemashkov-gw-01(config)# exit
msk-iemashkov-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-iemashkov-gw-01# configure terminal
msk-iemashkov-gw-01(config)# interface eth0
msk-iemashkov-gw-01(config-if)# ip address 172.16.20.1/25
msk-iemashkov-gw-01(config-if)# no shutdown
msk-iemashkov-gw-01(config-if)# exit
msk-iemashkov-gw-01(config)# interface eth1
msk-iemashkov-gw-01(config-if)# ip address 172.16.20.129/25
msk-iemashkov-gw-01(config-if)# no shutdown
msk-iemashkov-gw-01(config-if)# exit
msk-iemashkov-gw-01(config)# interface eth2
msk-iemashkov-gw-01(config-if)# ip address 64.100.1.1/24
msk-iemashkov-gw-01(config-if)# no shutdown
msk-iemashkov-gw-01(config-if)# exit
msk-iemashkov-gw-01(config)# write memory
% Unknown command: write memory
msk-iemashkov-gw-01(config)# exit
msk-iemashkov-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-iemashkov-gw-01#
```

Рис. 12: Настройка второго маршрутизатора на ipv6 адреса

Выполнение лабораторной работы

```
PC1-ssh@kali: ~$ ifconfig
vccs ping 172.16.20.138
64 bytes from 172.16.20.138: icmp_seq=1 ttl=63 time=0.314 ms
64 bytes from 172.16.20.138: icmp_seq=2 ttl=63 time=0.427 ms
64 bytes from 172.16.20.138: icmp_seq=3 ttl=63 time=1.446 ms
64 bytes from 172.16.20.138: icmp_seq=4 ttl=63 time=2.095 ms
64 bytes from 172.16.20.138: icmp_seq=5 ttl=63 time=1.386 ms
vccs ping 64.100.1.10
64 bytes from 64.100.1.10: icmp_seq=1 ttl=63 time=4.756 ms
64 bytes from 64.100.1.10: icmp_seq=2 ttl=63 time=2.403 ms
64 bytes from 64.100.1.10: icmp_seq=3 ttl=63 time=1.507 ms
64 bytes from 64.100.1.10: icmp_seq=4 ttl=63 time=2.279 ms
64 bytes from 64.100.1.10: icmp_seq=5 ttl=63 time=1.778 ms
vccs trace 64.100.1.10
Trace to 64.100.1.10: 8 hops max, press Ctrl+C to stop
 1 172.16.20.1 4.296 ms 1.263 ms 0.763 ms
 2 *64.100.1.10: 1.916 ms (ICMP type=3, code=3, Destination port unreachable)
vccs trace 172.16.20.138
Trace to 172.16.20.138: 8 hops max, press Ctrl+C to stop
 1 172.16.20.1 1.573 ms 0.589 ms 0.649 ms
 2 *172.16.20.138: 1.281 ms (ICMP type=3, code=3, Destination port unreachable)
vccs []

PC1-ssh@kali: ~$ ifconfig
vccs ping 172.16.20.10
64 bytes from 172.16.20.10: icmp_seq=1 ttl=63 time=0.180 ms
64 bytes from 172.16.20.10: icmp_seq=2 ttl=63 time=0.181 ms
64 bytes from 172.16.20.10: icmp_seq=3 ttl=63 time=0.192 ms
64 bytes from 172.16.20.10: icmp_seq=4 ttl=63 time=0.187 ms
64 bytes from 172.16.20.10: icmp_seq=5 ttl=63 time=0.202 ms
vccs trace 172.16.20.10
Trace to 172.16.20.10: 8 hops max, press Ctrl+C to stop
 1 172.16.20.129 1.718 ms 1.068 ms 0.538 ms
 2 *172.16.20.10: 1.448 ms (ICMP type=3, code=3, Destination port unreachable)
vccs ping 64.100.1.10
64 bytes from 64.100.1.10: icmp_seq=1 ttl=63 time=4.184 ms
64 bytes from 64.100.1.10: icmp_seq=2 ttl=63 time=1.627 ms
64 bytes from 64.100.1.10: icmp_seq=3 ttl=63 time=1.508 ms
64 bytes from 64.100.1.10: icmp_seq=4 ttl=63 time=2.184 ms
64 bytes from 64.100.1.10: icmp_seq=5 ttl=63 time=1.735 ms
vccs trace 64.100.1.10
Trace to 64.100.1.10: 8 hops max, press Ctrl+C to stop
 1 172.16.20.129 1.567 ms 0.731 ms 0.550 ms
 2 *64.100.1.10: 1.932 ms (ICMP type=3, code=3, Destination port unreachable)
vccs []
```

Рис. 13: Взаимный пинг и невозможность обращения к серверу по ipv4 адресу

В ходе данной лабораторной работы мной были изучены принципы распределения и приобретены навыки по настройке адресного пространства на устройствах сети.